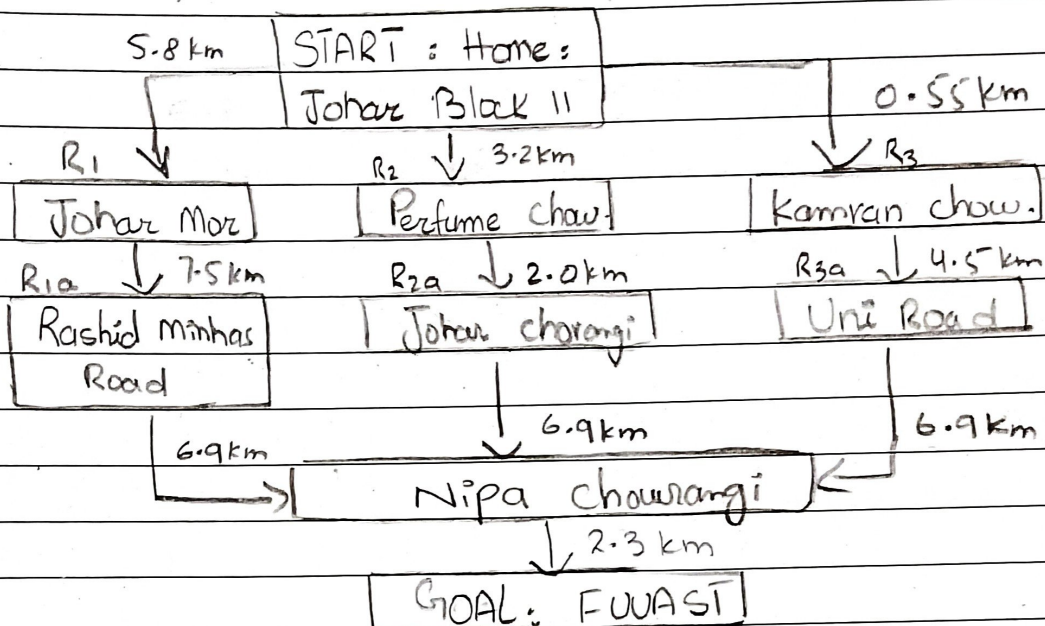


Dated: \_\_\_\_\_

Home To Uni

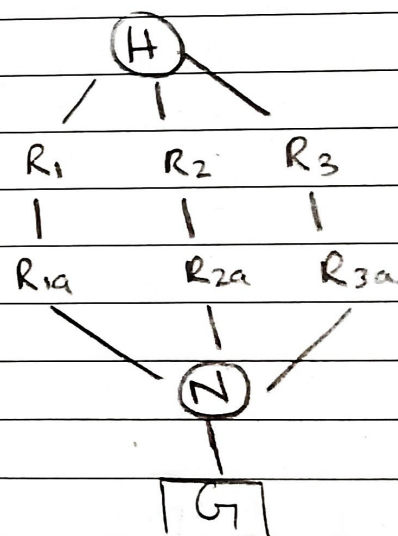
A\*



Route 1: 9.1 km

Route 2: 8.9 km

Route 3: 9.1 km



Connections

$H \rightarrow R_1, R_2, R_3$

$R_1 \rightarrow R_{1a}$

$R_2 \rightarrow R_{2a}$

$R_3 \rightarrow R_{3a}$

$R_{1a} \rightarrow N$

$R_{2a} \rightarrow N$

$R_{3a} \rightarrow N$

$N \rightarrow G$

Dated: \_\_\_\_\_

### Heuristic Values ( $h(n)$ )

$H \rightarrow 9.1, 8.9, 9.1$

$R_1 \rightarrow 5.8$

$R_2 \rightarrow 3.2$

$R_3 \rightarrow 0.55$

$R_{1a} \rightarrow 7.5$

$R_{2a} \rightarrow 2.0$

$R_{3a} \rightarrow 4.5$

$N \rightarrow 2.3$

$G \rightarrow 0$

Step: 0: Initialization

$PQ = [H]$

$V = \emptyset$

home  $\rightarrow$  None

| g | h   | f   |
|---|-----|-----|
| 0 | 8.9 | 8.9 |

Step: 1: Expand home:

$PQ = [R_1, R_2, R_3]$

$V = [H]$

| g | h    | f    | Nodes |
|---|------|------|-------|
| 1 | 5.8  | 6.8  | $R_1$ |
| 1 | 3.2  | 4.2  | $R_2$ |
| 1 | 0.55 | 1.55 | $R_3$ |

Choose  $R_3$

Parent Table update

$H \rightarrow$  None

$R_3 \rightarrow H$

Step: 2: Expand  $R_3$ :

$PQ = [R_{3a}]$

$V = [H, R_3]$

| g | h   | f   | Node     |
|---|-----|-----|----------|
| 2 | 4.5 | 6.5 | $R_{3a}$ |

Dated: \_\_\_\_\_

Parent Table update

H - None

$R_3 \rightarrow H$

$R_{3a} \rightarrow R_3$

Step = 3: Expand:  $R_{3a}$  :

| g | h   | t   | Node |
|---|-----|-----|------|
| 3 | 2-3 | 5-3 | N    |

PQ: [N]

V: [H,  $R_3$ ,  $R_{3a}$ ]

Parent Table update:

H - None

$R_3$  - H

$R_{3a}$  -  $R_3$

N -  $R_{3a}$

Step = 4: Expand N:

| g | h | t | Node |
|---|---|---|------|
| 4 | 0 | 4 | G    |

PQ: [G]

V: [H,  $R_3$ ,  $R_{3a}$ , N]

Parent Table update:

H - None

$R_3$  - H

$R_{3a}$  -  $R_3$

N -  $R_{3a}$

G - N

Dated: \_\_\_\_\_

- Start from goal & go backwards

$G \rightarrow N \rightarrow R3a \rightarrow R3 \rightarrow Home$

Here:  $G$  is goal (FVUAST)  
 $N$  is Nipa

Now Reverse it:

$Home \rightarrow R3 \rightarrow R3a \rightarrow N \rightarrow G$